

Mosaic Spend Intelligence GenAI Agent

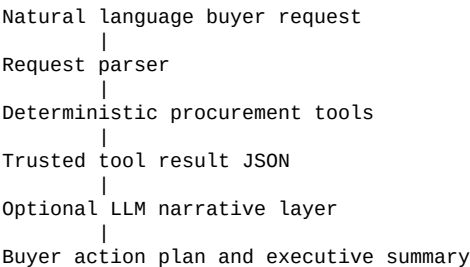
Build guide for an Aramark inspired procurement intelligence agent using synthetic purchase orders, supplier master data, SKU catalog data, contracts, risk signals, deterministic tools, and a GenAI style recommendation layer.

Important: this portfolio project uses synthetic data only. It is inspired by public business problems and does not use Aramark proprietary data.

1. Business Problem

Procurement teams need to harmonize purchase orders, supplier records, SKU catalogs, contract prices, and disruption signals. The agent identifies price leakage, maverick spend, supplier risk, single-source SKU risk, and substitution opportunities.

2. Agent Architecture



The GenAI layer does not calculate facts. The code computes spend, variance, maverick spend, risk scores, and substitution options. The LLM or offline narrative writer explains those trusted outputs.

3. Project Folder Structure

```
mosaic_spend_intelligence_genai_agent/
├── data/
│   ├── purchase_orders.csv
│   ├── suppliers.csv
│   ├── sku_catalog.csv
│   ├── contracts.csv
│   ├── disruptions.csv
│   └── procurement_policy.md
├── src/
│   ├── tools.py
│   ├── agent.py
│   └── run_demo.py
├── output/
│   ├── demo_recommendation.json
│   └── demo_recommendation.md
├── README.md
└── requirements.txt
```

4. Data Tables

Table	Purpose
purchase_orders.csv	Synthetic line level purchase order spend by SKU, supplier, location, quantity, price, and spend.
suppliers.csv	Supplier reliability, region, diversity status, sustainability score, and risk status.
sku_catalog.csv	SKU category, substitution group, unit, and criticality.
contracts.csv	Contract price and preferred supplier mapping.

5. Coding Part Explained

tools.py

This file contains deterministic procurement tools. These functions calculate the facts used by the agent.

```
enrich_purchase_orders()
- Joins purchase orders to SKU, supplier, and contract data.
- Calculates price_variance_pct.
- Calculates potential_savings.

detect_price_variance()
- Flags lines more than threshold above contract price.

detect_maverick_spend()
- Flags non-preferred supplier purchases.

supplier_scorecard()
- Calculates supplier risk using reliability, sustainability, active disruptions, and risk status.

build_supplier_sku_graph()
- Builds a supplier-SKU network using NetworkX.

recommend_alternates_for_sku()
- Finds alternate SKUs and suppliers in the same substitution group.
```

agent.py

This file wraps the tools into a procurement copilot workflow.

```
MosaicSpendIntelligenceAgent
  parse_request()
    Converts user text into structured constraints.
  apply_constraints()
    Filters data by category or location.
  run_tools()
    Runs spend, variance, maverick, risk, graph, and substitution tools.
  generate_prompt()
    Creates a grounded prompt for an optional LLM.
  deterministic_narrative()
    Produces an offline GenAI-style buyer summary.
  run()
    Executes the full agent workflow.
```

6. How to Run

```
cd mosaic_spend_intelligence_genai_agent
pip install -r requirements.txt
python src/run_demo.py
```

The demo writes output/demo_recommendation.json and output/demo_recommendation.md.

7. Demo Request

Analyze Protein spend for price variance over 20%, maverick spend, supplier risk, and substitution opportunities. Prefer reliable suppliers and include diverse supplier opportunities when comparable.

8. Demo Output Summary

The demo identifies total synthetic spend, potential savings, maverick spend, diverse supplier spend, top price variance issue, highest supplier risk, and a substitution recommendation. Example: replace or supplement a high-cost Chicken Breast purchase with Turkey Strips from a more reliable supplier when appropriate.

9. Interview Explanation

I built a Mosaic inspired GenAI procurement agent that analyzes synthetic purchase orders, supplier master data, SKU catalogs, contract prices, and disruption signals. The system flags maverick spend, contract price leakage, supplier risk, and single-source SKU exposure. The LLM is not calculating the facts. The deterministic tools calculate the trusted procurement metrics, and the GenAI layer explains them as a buyer action plan. This keeps the system auditable and enterprise ready.

10. Guardrails

- Do not invent supplier availability.
- Do not invent contract prices.
- Do not recommend suppliers that are not in the supplier data.
- Do not present synthetic savings as real company savings.
- Always label outputs as synthetic and public-problem inspired.

11. Production Architecture Extension

In production this would map to Azure Data Factory for ingestion, Databricks or Synapse for medallion lakehouse processing, Delta tables for Bronze/Silver/Gold layers, Unity Catalog for governance, MLflow for risk and savings model tracking, vector search for procurement policies, and Power BI for procurement dashboards.

12. 48-Hour Learning Tasks

- Run the demo and inspect demo_recommendation.json.
- Open tools.py and understand enrich_purchase_orders first.
- Trace price variance calculation by hand for one PO line.
- Run detect_maverick_spend and compare preferred_supplier values.
- Open agent.py and trace run() from request to final summary.
- Modify the request to Analyze Produce spend and re-run.
- Add one new supplier and one new contract to see how recommendations change.